

Tommaso Guerrini

LEAD DATA SCIENTIST | FINANCIAL ML | TABULAR AI | EXPLAINABLE AI

Milano, Italy | Substack: @thetabularguy

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- **Lead Data Scientist & Tabular AI Specialist** with 10 years of experience building production ML systems, from credit-risk early warning to tabular foundation models, across FinTech, insurance, and consulting, plus industrial predictive-maintenance R&D.
- Author of **The Tabular Guy** on Substack and technical portfolio at tomwarrens.github.io.
- **3rd worldwide** in the Prior Labs TabPFN World Cup competition (2026); Kaggle Notebook Master with a career high of **#1 in Italy** and **#163 worldwide**.
- Specialized in credit-risk PD and early-warning modeling, tabular foundation models (TabICL, TabPFN), SHAP and surrogate explainability, temporal validation design, unsupervised anomaly detection, and hierarchical time-series forecasting.
- Bridges deep statistical modeling with production MLOps/GitOps delivery (Kubernetes, KubeFlow, ArgoCD, KServe, MLflow) and product integration (FastAPI, React) to deliver validated, commercial AI products.

Selected Impact

- **Bank-Validated Early Warning:** In the bank's own forward test, matched the incumbent model's Stage 3 capture with **~4x fewer** flagged loans across **400M+ EUR** of SME exposure (~7k counterparties).
- **Product Ownership:** Built and led the early-warning credit product behind **~500k EUR ARR**, **~80%** of team revenue.
- **Default Discrimination:** Delivered a **~21%** relative AUC gain over the bank's internal risk indicators, flagging the highest-risk decile **~5 months** ahead of default.
- **Foundation Model in Production:** Productionized TabICL v2 (open-weight) as champion PD model, replacing GBDT and TabPFN, with retrieved-context inference **4.3x** faster than full context.
- **Alarm Calibration Without Labels:** Showed that a CNC manufacturer's reference predictive-maintenance method set alarm thresholds producing a **28.6%** median false-alarm rate across **19** machines against a 5% target; replaced them with held-out temporal calibration.

Experience

CardoAI

Milano, Italy (Remote)

HEAD OF TABULAR AI / LEAD DATA SCIENTIST

Jul. 2022 – Present

- Owned modeling and product integration for doValue's early-warning credit suite (loan-tape ingestion, PD scoring, SHAP risk drivers, FastAPI/React monitoring dashboard), live on **400M+ EUR** of SME exposure at an Italian commercial bank (Banca Sella).
- Developed three PD model generations on longitudinal SME loan tapes (GBDT bootstrapped on European DataWarehouse data, TabPFN v2.6, TabICL v2) under expanding temporal validation with a 6-month embargo, reaching **AUC ~0.93** vs 0.77 for the bank's internal indicator.
- Productionized TabICL v2 (BSD-3) on a new scoring endpoint; KNN retrieved-context inference gained **+0.0147** never-in-arrears AUC (loans with no arrears history) over TabPFN across four temporal holdouts, running **4.3x** faster than full context.
- Architected a two-tier explainability engine (exact permutation SHAP on latest snapshots, LightGBM surrogate on TabICL log-odds, R^2 0.98–0.997), cutting **~2,000** estimated compute hours to a **2h10m** build.
- Optimized TabICL serving on Nvidia T4 GPUs through KV-cache pinning and batching, cutting per-row SHAP latency from 10.9s to 0.22s (**~50x**), and patched a silent 0.0076 PD drift introduced by batching.
- Partnered with DevOps to operationalize models on Kubernetes, KubeFlow, ArgoCD, KServe, and MLflow with automated CI/CD and audit trails; automated data profiling cut client onboarding time by **~30%**.
- Led two PhD data scientists, ran hiring interviews, supported presales through client workshops and demos, and defined ML standards across validation, explainability, stability monitoring, and governance.

Tech: TabICL, TabPFN, LightGBM, XGBoost, SHAP, DeepChecks, MLflow, Airflow, Databricks, PySpark, AWS, Kubernetes, KServe, FastAPI, React.

Emmegi

Milano, Italy (Remote)

FREELANCE DATA SCIENTIST – PREDICTIVE MAINTENANCE

Apr. 2026 – Present

- Built an unsupervised predictive-maintenance pipeline for CNC machining centres (no fault labels) on raw PLC servo data from **19** end-of-line tests; identified friction torque at matched speed and direction as a stable test fingerprint (split-half rank correlation **0.98**).
- Showed that the reference method's i.i.d. bootstrap thresholds produced a **28.6%** median false-alarm rate across 19 machines against a 5% target, and replaced them with held-out temporal calibration.
- Designed a label-free evaluation harness (synthetic degradations, temporal splits, held-out machines) to benchmark Isolation Forest and autoencoders against statistical detectors; caught **3 of 4** real failures at **1.7%** false alarms on public MetroPT-3 data.

Tech: Python, PyTorch, scikit-learn, TSFresh, SciPy, statsmodels, pandas, PyArrow, pytest.

Generali Italia

Milan, Italy

MACHINE LEARNING SCIENTIST / DATA SCIENTIST

Jan. 2020 – Jun. 2022

- Led the motor-claims fraud workstream as technical lead: took a tabular fraud model for claim triage from feature engineering to integration in the live decision workflow, and ran a Neo4J graph-feature pilot on a multi-million-node entity graph.
- Developed and deployed several claim-document classifiers (classical NLP features with gradient boosting) as microservices on GCP Cloud Functions, App Engine, and BigQuery.
- Architected a versioned PySpark feature store orchestrated with Jenkins and Airflow to feed fraud pipelines, and introduced CI/CD templates and evaluation metrics reused across the innovation department.

Tech: Google Cloud Platform (Vertex AI, BigQuery, Cloud Functions), PySpark, LightGBM, Optuna, Neo4J, Docker, Airflow, Jenkins.

Data Reply IT

Milan, Italy

DATA SCIENTIST

Aug. 2016 – Dec. 2019

- Developed hierarchical demand forecasting for Pirelli's supply chain using hts (Hyndman) in R with ARIMA and GBDT base learners, and classical time-series forecasting for Eni.
- Engineered real-time TV audience forecasting for Mediaset (LightGBM, dense temporal features), improving baseline accuracy by **>10%** and spike-period accuracy by **~25%**.
- Built churn prediction and large-scale PySpark ETL for Vodafone marketing campaigns, and migrated on-premise analytics to GCP and AWS (EMR, Athena, Glue).

Tech: Python, PySpark, LightGBM, XGBoost, AWS (EMR, Athena, Glue), Google Cloud Platform (BigQuery), Git, Agile.

Certifications and Awards

- **3rd Place – Prior Labs TabPFN World Cup Competition** (2026): Individual entry built on fixture-specific TabPFN context. [\[Details\]](#)
- **Author – The Tabular Guy** (Jun. 2026 – Present): Substack on tabular foundation models (TabPFN, TabICL) in practice. [\[Substack\]](#)
- **Kaggle Notebook Master** (Jul. 2021 – Present): Career high of **#1 in Italy** and **#163 worldwide**. [\[Kaggle Profile\]](#)
- **Winner – CLADAG / TIM Data Science Competition** (Oct. 2019): 1st place nationally in a churn classification challenge.

Core Skills

- **Machine Learning & Data Science:** Credit-risk PD and early-warning modeling, tabular foundation models (TabICL, TabPFN), SHAP and surrogate explainability, temporal validation design (expanding windows, embargo, label maturity), time-series and hierarchical forecasting, unsupervised anomaly detection, statistical process monitoring, model validation, drift monitoring, survival analysis.
- **Modeling Frameworks:** LightGBM, XGBoost, scikit-learn, TabICL, TabPFN, SHAP, PyTorch, TSFresh, DeepChecks, Optuna, MLflow.
- **Data Engineering, MLOps & Cloud:** Python, SQL, PySpark, Airflow, Kubernetes, Kubeflow, KServe, ArgoCD, Docker, Jenkins, Git, CI/CD, feature stores, FastAPI, React, Dash, GPU inference (Nvidia T4), AWS (EMR, Athena, Glue, S3, EC2), Google Cloud Platform (BigQuery, Vertex AI, Cloud Functions, Dataproc), Databricks, Neo4J.
- **Leadership & Commercial Impact:** Team lead of two PhDs, hiring interviews, ML standards and mentoring, product ownership, technical presales and client workshops, client-facing technical reports, Agile delivery (Jira).

Talks, Teaching and External Recognition

- **European DataWarehouse Irish Securitisation Event – Dublin** (2026): Keynote on AI, data quality, and automation in securitisation.
- **MSCA Digital Finance – Machine Learning in Industry** (2026): Course instructor on applied ML and financial ML productization.

Education

Politecnico di Milano

Milan, Italy

M.Sc. IN MATHEMATICAL ENGINEERING, APPLIED STATISTICS (110/110)

Oct. 2014 – Apr. 2017

- Final Thesis: “Machine Learning Models for Forecasting of Daily Supermarket Sales” (Demand forecasting & retail elasticity).
- Relevant Coursework: Applied Statistics, Machine Learning, Parallel Computing, Bayesian Statistics.

Politecnico di Milano

Milan, Italy

B.Sc. IN PHYSICS ENGINEERING (95/110)

Oct. 2011 – Sep. 2014

- Relevant Coursework: Partial Differential Equations, Quantum Physics, Linear Algebra, Mathematical Analysis.